

AI Dataset Cleaning & Quality Report

Dataset File: [dirty_test_data.csv](#)

Session ID:	f49c3a6e-5b8e-4ed4-b218-9a8fef449964
Quality Score Before:	89/100
Quality Score After:	98/100
Execution Time:	2026-07-26 22:19:59

Confidentiality & Compliance Notice

This report outlines automated data cleaning operations executing on proprietary files. All code transformations, parsing summaries, and metrics modifications are logged for audit purposes.

Executive Summary

Successfully executed cleaning workflows against dataset 'dirty_test_data.csv'. The overall dataset quality score improved from 89/100 to 98/100.

Clean Metrics Comparison

Metric	Before Cleaning	After Cleaning	Variance
Total Rows	6	5	+16.7%
Total Columns	6	5	+16.7%
Missing Values	3	0	+100.0%
Duplicate Rows	1	0	+100.0%
Quality Score	89	98	+9.0%
Memory Footprint	1.58 KB	0.93 KB	+41.0%

Dataset Quality Inspection Detail

Column Target	Defect / Finding Description	Proposed Mitigation Recommendation
date_registered	Contains 1 missing values (16.7%).	Impute missing values using column median or mean.
salary	Contains 1 missing values (16.7%).	Impute missing values using column median or mean.
active	Contains 1 missing values (16.7%).	Impute missing values using column median or mean.
Global Schema	Found 1 identical duplicate rows.	Drop duplicate rows keeping the first occurrence.
name	1 values contain leading or trailing spaces.	Trim whitespace padding.
email	Detected 3 values with invalid email formats.	Validate syntax regulations or discard corrupt email addresses.
salary	Contains 1 outlier records outside IQR range boundaries.	Clamp outliers or drop extreme values.

Cleaning Log & Actions Executed

Cleaner Executed	Operation Parameters	Rows/Cells Affected
Duplicate Cleaner	{'keep': 'first'}	1
Text Cleaner	{'trim_spaces': True, 'case': 'none', 'remove_extra_spaces': True}	2
Validator	{'email_columns': [], 'phone_columns': []}	2
Datetime Cleaner	{'output_format': '%Y-%m-%d'}	3
Outlier Cleaner	{'strategy': 'clamp'}	1
Missing Value Cleaner	{'strategy': 'median'}	8
Datatype Cleaner	{'downcast_numeric': True, 'normalize_booleans': True, 'category_conversion': True}	2

Suggestions for Upstream Data Collection

- Set schema validation on upstream datastore triggers.
- Sanitize leading and trailing whitespace strings in input forms.